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A small narrow tube, **oesophagus** lies in the 5th segment and continues into a muscular **gizzard** in the 6th segment. The gizzard helps in the grinding of soil particles and decaying leaves. **Intestine** starts from the 7th segment and continues till the last segment. The dorsal wall of the intestine is folded into the cavity as the **typhlosole**. This fold contains blood vessels and increases the absorptive area of the intestine. The inner epithelium consists of columnar cells and glandular cells. The alimentary canal opens to the exterior through the anus.

Intestinal Caeca

In *Metaphire posthuma*, the 26th segment has a pair of short conical out growths called intestinal caecae. It is extended anteriorly up to the 22nd segment. These are digestive glands and secrete an amylolytic enzyme for the digestion of starch. Intestinal caecae are not present in many species of earthworms such as the *Lampito mauritii*.

The ingested organic rich soil passes through the digestive tract where digestive enzymes breakdown complex food into smaller absorbable units. The simpler molecules are absorbed through the intestinal membrane and are utilized. The undigested particles along with earth are passed out through the anus, as **worm castings** or **vermicasts**. The pharyngeal or salivary gland cells and the glandular cells of the intestine are supposed to be the digestive glands which secrete digestive enzymes for digestion of food.

Earthworms have "setae", which are small hair-like bristles, though they are not composed of the same material as human hair. The setae help the earthworm to anchor itself while feeding or mating. It is also found in the wings/exoskeletons of insects. What biological matter forms these structures?

Respiratory System

The earthworm has no special respiratory organs like lungs or gills. Respiration takes place through the body wall. The outer surface of the skin is richly supplied with blood capillaries which aid in the diffusion of gases. Oxygen diffuses through the skin into the blood while carbon dioxide from the blood diffuses out. The skin is kept moist by mucous and coelomic fluid and facilitates exchange of gases.

Circulatory system

Lampito mauritii exhibits a closed type of blood vascular system consisting of blood vessels, capillaries and lateral hearts (Figure 4.4). Two median longitudinal vessels run above and below the alimentary canal as dorsal and ventral vessels of the earthworm. There are paired valves in the dorsal vessels which prevent the backward flow of the blood. The ventral vessel has no valves and is non contractile, allowing the backward flow of blood. In the anterior part of the body the dorsal vessel is connected with the ventral vessel by eight pairs of **commissural vessels** or the **lateral hearts** lying in the 6th to 13th segments. These vessels run on either side of the alimentary canal and pump blood from the dorsal vessel to the ventral